LEVEL 13 · PART VI

Databases & Distributed Systems

13

Goal: Reason about data stores, messaging and distributed-systems trade-offs when operating production systems.

RECOMMENDED TIME
4-6 weeks

TOPICS
15

PRACTICE / CAPSTONE
Run Postgres primary+replica, Redis and Kafka (or RabbitMQ) locally; demonstrate replication lag, an idempotent consumer and a PITR restore.

Topics

#	Topic	What you will learn
1	PostgreSQL Indexes **p.2579**	B-tree/GIN/partial, EXPLAIN ANALYZE, index cost
2	Transactions and Locks **p.2593**	ACID, isolation levels, row/table locks, deadlocks
3	MVCC **p.2607**	snapshots, xmin/xmax, vacuum, bloat, long-running transaction danger
4	Replication **p.2623**	streaming replication, sync vs async, replicas, lag, failover tools
5	Backups and Point-in-Time Recovery **p.2639**	pg_dump vs base backup + WAL, PITR, pgBackRest/WAL-G, restore drills
6	Database Connection Pooling **p.2655**	PgBouncer modes, max_connections, pool sizing math
7	Redis **p.2669**	data structures, caching patterns, persistence RDB/AOF, eviction, Sentinel/Cluster
8	Kafka **p.2685**	topics, partitions, consumer groups, offsets, retention, ordering
9	RabbitMQ **p.2701**	exchanges, queues, acks, DLQ, vs Kafka
10	Horizontal and Vertical Scaling **p.2715**	when each, stateless vs stateful, read replicas

#	Topic	What you will learn
11	Partitioning and Sharding **p.2727**	table partitioning, shard keys, hotspots, resharding
12	Consistency, Availability and the CAP Theorem **p.2739**	CAP/PACELC, linearizability, real systems classified
13	Eventual Consistency **p.2751**	convergence, read-your-writes, conflict resolution, CRDTs intro
14	Idempotency **p.2763**	idempotency keys, at-least-once delivery, dedup, retries
15	Distributed Locks and Queues **p.2777**	Redis locks, fencing tokens, leases, work queues, outbox pattern
★	Level capstone project **p.2793**	Run Postgres primary+replica, Redis and Kafka (or RabbitMQ) locally; demonstrate replication lag, an idempotent consumer and a PITR restore.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

